MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
THIRD SEMESTER
Discipline Specific Elective course (DSE B) –MG3DSEBBT202
PLANT AND ANIMAL PHYSIOLOGY
(2024 ADMISSION ONWARDS)

Duration : 2 Hours **Maximum Marks : 70**

Remember (K), Understanding (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skills (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

One Word Answer
Answer any **10**
Each question carries **2** marks

1.The chronic condition characterized by inflammation of the airways [K] / [CO 1]
2.Name the organs present in the dorsal body cavity. [K] / [CO 1]
3.The location of the adrenal glands with reference to the kidneys. [U] / [CO 1]
4. List out three stress hormones in plants. [K] / [CO 3]
5.List out the two essential processes that work together to breakdown food into usable nutrients and move them to blood stream. [K] / [CO 1]
6. What is known for the relative volume of formed elements in the blood? [K] / [CO 1]
7.The condition where plaque builds up in the arteries, leading to heart disease. [K] / [CO 1]
8.Plant hormone which promotes cell division and cell elongation. [U] / [CO 4]
9.Name any two treatment methods for neurological disorders. [K] [CO 2]
10.Expand EEG. [K] / [CO 2]
11.Name the environmental factors affecting plants. [U] / [CO 3]
12.What are the crop improvement methods? [K] / [CO 4]

(2 x 10 = 20)

PART B

Short Essay
Answer any **4** questions
Each question carries**5** marks

13. Explain the mechanism of transportation in plants. [U] / [CO 3]
14. How does the body regulate blood glucose level? [K] / [CO 1]
15. Discuss on the typesof essential elements for plants. [U] / [CO 3]
16. How do white blood cells help the body to fight infections? [K] / [CO 1]
17. Describe on how the human physiology helps in the diagnosis of diseases. [U[/ [CO2]
18. Explain the defense mechanism in plants. [U] / [CO 4]

(5 x 4 = 20)

PART C

Essay

Answer any **2** questions

Each question carries **15** marks

19. Compare and contrast the mechanisms of photoperiodism and vernalization and the response of hormones in plants. [U]/[CO3]

20. Summarize how the plant physiology helps in crop improvement and plant breeding [U] /[CO 4]

21. Interpret the role of neurotransmitters in transmitting signals across synapses[U] /[CO 1]

22. Describe the different types of plant movements. [R] / [CO3]

(2 x 15 =30)